**Computational Fluid Dynamics**
**EN 530.767**
**HW 1**

Weixuan Li

# 1. Problem 1

## 1.1 Objective

Explore the finite-difference solution to the 1-D wave equation.

$$u_t + u_x = 0 \text{ on } 0 \le x \le 2\pi$$

With periodic boundary condition and initial condition $u(0,t) = u(2\pi, t)$.

## 1.2 Methodology

To solve it numerically, the first step is to discretize the equation with an explicit scheme. The time is discretized by $\Delta t = 0.001$ and the space is discretized by $\Delta x = \frac{2\pi}{20}$. Four upwind spatial discretization schemes are used, which can be derived from Taylor table shown in Figure 1.

| | $u_j$ | $u_j'$ | $u_j''$ | $u_j'''$ |
|---|---|---|---|---|
| $a_0 u_j$ | $a_0$ | $0$ | $0$ | $0$ |
| $a_1 u_{j+1}$ | $a_1$ | $a_1 \Delta x$ | $a_1 \frac{\Delta x^2}{2}$ | $a_1 \frac{\Delta x^3}{6}$ |
| $a_2 u_{j-1}$ | $a_2$ | $-a_2 \Delta x$ | $a_2 \frac{\Delta x^2}{2}$ | $-a_2 \frac{\Delta x^3}{6}$ |
| $a_3 u_{j-2}$ | $a_3$ | $-2a_3 \Delta x$ | $2a_3 \Delta x^2$ | $-\frac{4}{3} a_3 \Delta x^3$ |
| $a_4 u_{j-3}$ | $a_4$ | $-3a_4 \Delta x$ | $\frac{9}{2} a_4 \Delta x^2$ | $-\frac{9}{2} a_4 \Delta x^3$ |

Figure 1 Taylor table used to derive the upwind schemes

From Taylor table, we can derive the discretized formula for the four schemes:

a. First-order upwind scheme

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{u_i^n - u_{i-1}^n}{\Delta x}$$

b. Second-order upwind scheme

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{u_{i-2}^n - 4u_{i-1}^n + 3u_i^n}{2\Delta x}$$

c. Third-order upwind scheme

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{-2u_{i-3}^n + 9u_{i-2}^n - 18u_{i-1}^n + 11u_i^n}{6\Delta x}$$

d. Third-order upwind biased scheme

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{u_{i-2}^n - 6u_{i-1}^n + 3u_i^n + 2u_{i+1}^n}{6\Delta x}$$

In order to compare the numerical results with the exact solution, the characteristic line method is used to derive the exact solution of the PDE.

First, select the characteristic line:

$$\frac{dx}{dt} = 1 \implies x = t + C \ (C \ is \ constant)$$

Then,

$$u_t + u_x = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\frac{dx}{dt} = \frac{du}{dt} = 0$$

Therefore, $u$ is a function of $C$, so that:

$$u(x,t) = f(x,t)$$

From the initial condition,

$$u(x,0) = f(x) = \sin(mx)$$

Therefore, $u$ can be determined as

$$u(x,t) = sin\,[m(x-t)]$$

This equation also satisfies periodic boundary condition. Thus, the exact solution is:

$$u(x,t) = sin\,[m(x-t)]$$

**1.3 Results and Discussion**

To better understand the behaviors of the numerical solutions, the modified wave numbers of the four schemes are derived (shown in Figure 2), and their curves along with the exact solution wave number curve are plotted in Figure 3.

$$u_i = \hat{u} e^{ikx}$$

$$\frac{\partial u}{\partial t} = -\frac{\partial u}{\partial x} = -ik\hat{u}e^{ikx} = -iku_i$$

This $k$ is the wave number of exact solution

For the upwind schemes,

a. $1^{st}$ order upwind

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{u_i^n - u_{i-1}^n}{\Delta x} = -ik' u_i$$

$$= -u_i \frac{1 - e^{-ik\Delta x}}{\Delta x} = -ik' u_i$$

$$\Rightarrow ik' = \frac{1 - e^{ik\Delta x}}{\Delta x} \Rightarrow k' = \frac{\sin(k\Delta x)}{\Delta x} + i\frac{\cos(k\Delta x) - 1}{\Delta x}$$

b. $2^{nd}$ order upwind

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{u_{i-2} - 4u_{i-1} + 3u_i}{2\Delta x} = -ik' u_i$$

$$\frac{\cos(2k\Delta x) - i\sin(2k\Delta x)}{2\Delta x} - \frac{2[\cos(k\Delta x) - i\sin(k\Delta x)]}{\Delta x} + \frac{3}{2\Delta x} = ik'$$

$$\Rightarrow k' = -\frac{\sin(2k\Delta x)}{2\Delta x} + \frac{2\sin(k\Delta x)}{\Delta x} + i\left[-\frac{\cos(2k\Delta x)}{2\Delta x} + \frac{2\cos(k\Delta x)}{\Delta x} - \frac{3}{2\Delta x}\right]$$

c. $3^{rd}$ order upwind

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{11u_i - 18u_{i-1} + 9u_{i-2} - 2u_{i-3}}{6\Delta x} = -ik' u_i$$

$$\Rightarrow ik' = \frac{11 - 18e^{-ik\Delta x} + 9e^{-i2k\Delta x} - 2e^{-i3k\Delta x}}{6\Delta x}$$

$$\Rightarrow k' = \frac{18\sin(k\Delta x) - 9\sin(2k\Delta x) + 2\sin(3k\Delta x)}{6\Delta x} + i\frac{-11 + 18\cos(k\Delta x) - 9\cos(2k\Delta x) + 2\cos(3k\Delta x)}{6\Delta x}$$

d. $3^{rd}$ order upwind biased

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{2u_{i+1} + 3u_i - 6u_{i-1} + u_{i-2}}{6\Delta x} = -ik' u$$

$$\Rightarrow ik' = u_i \frac{2e^{ik\Delta x} + 3 - 6e^{-ik\Delta x} + e^{-i2k\Delta x}}{6\Delta x}$$

$$\Rightarrow k' = \frac{8\sin(k\Delta x) - \sin(2k\Delta x)}{6\Delta x} + i\frac{-3 + 4\cos(k\Delta x) - \cos(2k\Delta x)}{6\Delta x}$$

Figure 2 Derivation of the modified numbers of the four upwind schemes

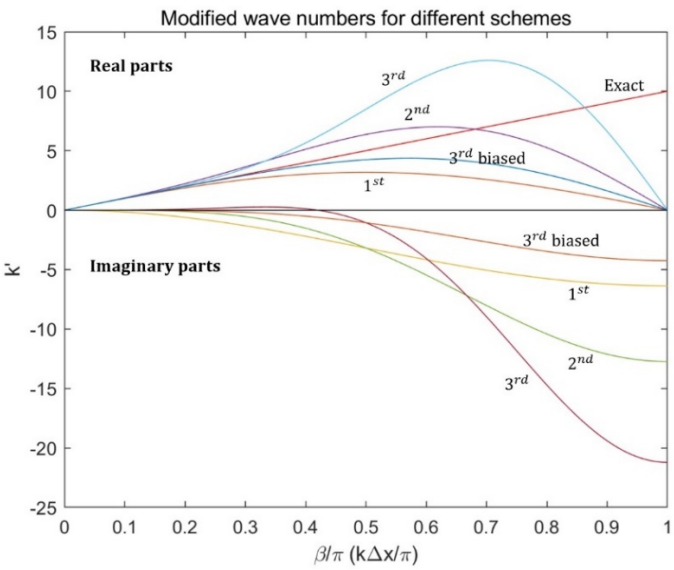
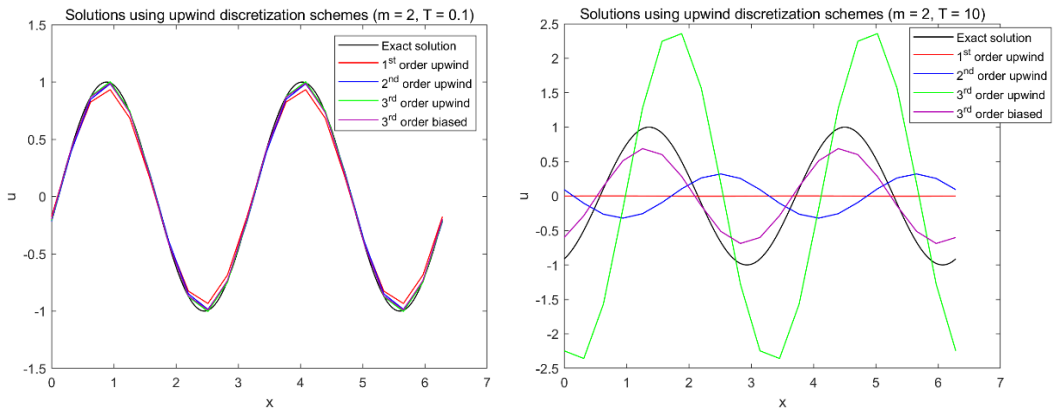


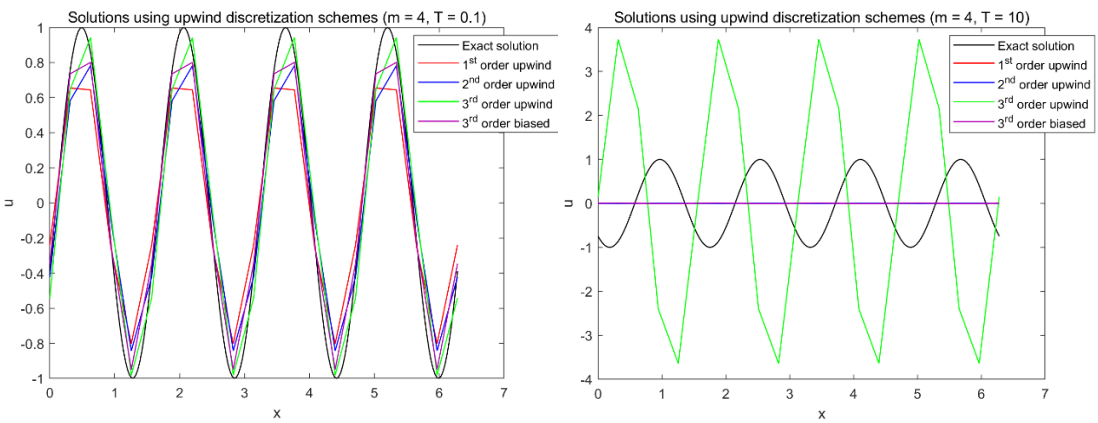

Figure 3 Modified wave number curves of the four upwind schemes and the exact solution

The numerical solutions of four schemes when $m = 2, 4, 6, 8$ at time $T = 0.1, T = 10$ are shown in Figure 4-7.



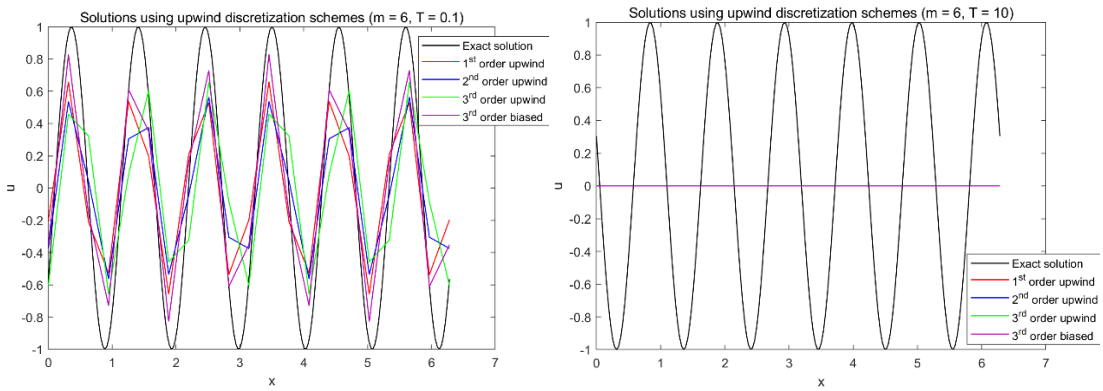

(a) (b)

Figure 4 Numerical solutions at $m = 2$ and $T = 0.1$ (a), $T = 10$ (b)



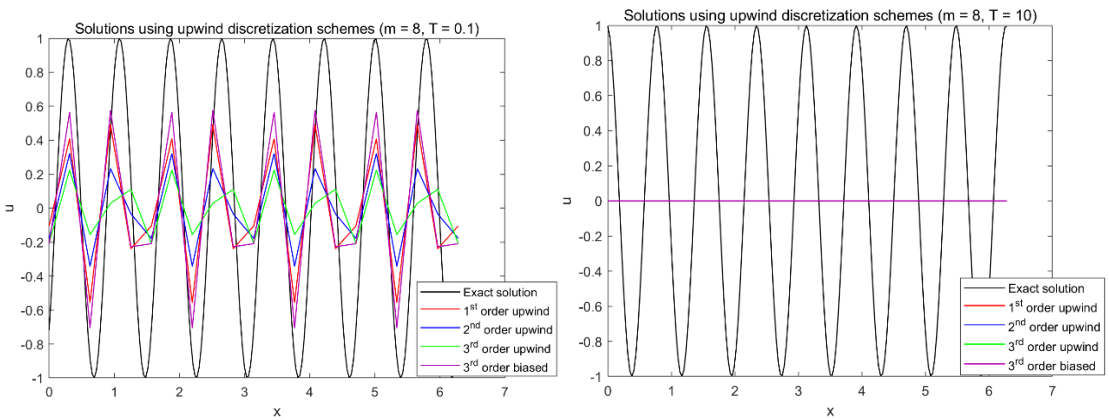
(a) (b)

Figure 5 Numerical solutions at $m = 4$ and $T = 0.1$ (a), $T = 10$ (b)




(a) (b)

Figure 6 Numerical solutions at $m = 6$ and $T = 0.1$ (a), $T = 10$ (b)




(a) (b)

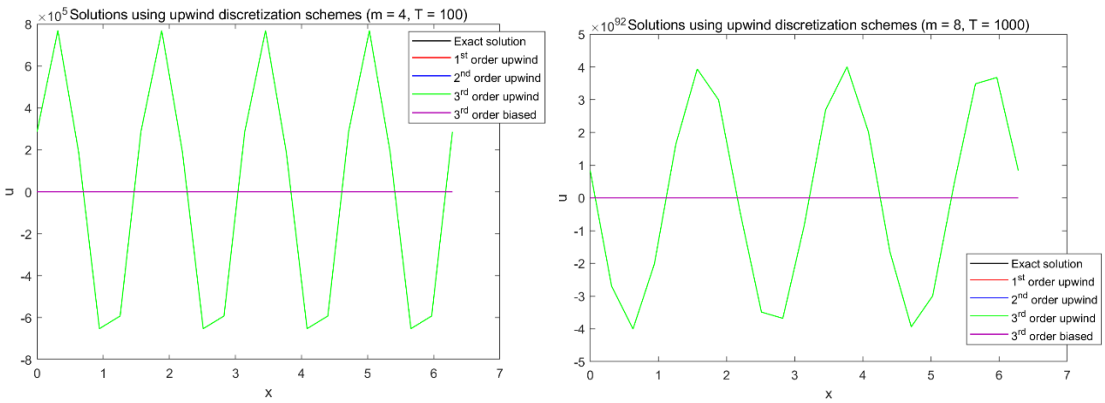
Figure 7 Numerical solutions at $m = 8$ and $T = 0.1$ (a), $T = 10$ (b)

At early time steps, this scheme performs well, as evidenced by the close alignment of the numerical solutions with the exact solution. However, with the progression of time integration, the differences become more pronounced. This is attributed to the differences in the real and imaginary parts between the exact wave number and the modified wave numbers.

The real parts represent wave speed. When β is small, the real parts of each scheme closely match both each other and the wave number curve of the exact solution. However, as β increases, the differences become more pronounced. As a result, these four schemes exhibit a phase difference relative to the exact solution, which intensifies over time. Figure 4 illustrates this phenomenon: at T = 0.1, there is no significant phase difference. But at T = 10, due to the variance in wave speeds, these four schemes exhibit a noticeable phase difference compared to the exact solution. Moreover, when m = 2, which corresponds to β = 0.2, the phase difference in the second-order upwind scheme

is most substantial. These results are in close agreement with the observations from Figure 3.

The magnitude of the imaginary part determines the amplification of each step in the numerical solution. From Figure 3, the imaginary parts of the modified wave number of four schemes are less than zero, except for the section between 0.2 and 0.4 in the curve of the third-order upwind scheme, where the imaginary part is slightly greater than zero. Therefore, as observed from Figures 4-7, with the progression of time, the numerical solutions of the four schemes gradually decay and approach to zero. However, as shown in Figures 4(b) and 5(b), when m = 2 and m = 4, corresponding to β = 0.2 and β = 0.4 respectively, the numerical solutions of the third-order upwind scheme diverge. This divergence occurs because in these cases, the imaginary part of the modified wave number is greater than zero, leading to an increasing magnitude of the numerical solution over time. While the other imaginary parts of the modified waver number of the third-order upwind scheme are less than zero, the numerical solution will also diverge given sufficient time. This result is illustrated in Figure 8, showcasing the numerical solutions of the third-order upwind scheme for the cases of m=4 at T=100, and m=8 at T=1000. These solutions diverge due to the round-off error in the sections where the imaginary part is greater than zero.




(a) (b)

Figure 8 Numerical solutions at an extreme long time $T = 1000$ when $m = 6$ (a), $m = 8$ (b)

## 1.4 Conclusion

In exploring the finite-difference solution to the 1-D wave equation, this study utilized four upwind spatial discretization schemes to solve the equation. At initial time step, these schemes align closely with the exact solution, indicating their effectiveness. However, as time progresses, differences become pronounced, due to the disparity in the real and imaginary parts of the wave numbers. The real parts, indicative of wave speed, affect the phase difference. Additionally, the imaginary parts magnitude influences the solution's amplification; negative values lead to decay, while positive values in the third-order upwind scheme result in divergence over time. These findings highlight the criticality of selecting appropriate discretization parameters and the impact of modified wave number characteristics on the accuracy of numerical solutions in solving 1-D wave equation.

## 2. Problem 2

### 2.1 Objective

Examine the aliasing error in solutions of two time-dependent differential equations:

$$u_t + u^2 = 0 \text{ and } u_t + \sin(3x)u = 0$$

With initial condition $u(x) = sin(x) + 0.5sin(4x)$ in a domain of size $2\pi$.

### 2.2 Methodology

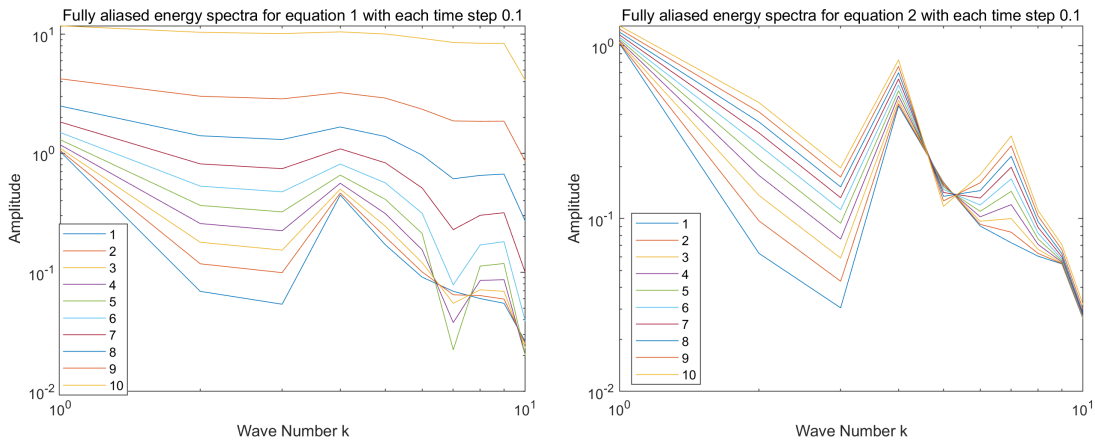
Implement the Forward Euler scheme to solve these two equations:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + (u_i^n)^2 = 0$$
$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \sin(3x_i)u_i^n = 0$$

For the fully aliased scheme, the space discretization $\Delta x = \frac{2\pi}{20}$ and the time discretization $\Delta t = 0.10$ are implemented. For the unaliased scheme, the space discretization $\Delta x = \frac{2\pi}{2000}$ is implemented with the time discretization $\Delta t = 0.01$ instead considering the convergence requirement. The fully aliased scheme is also implemented with the finer time discretization $\Delta t = 0.01$ to compare with the unaliased scheme. FFT is used to obtain the energy spectra at different wave numbers.

### 2.3 Results and Discussion

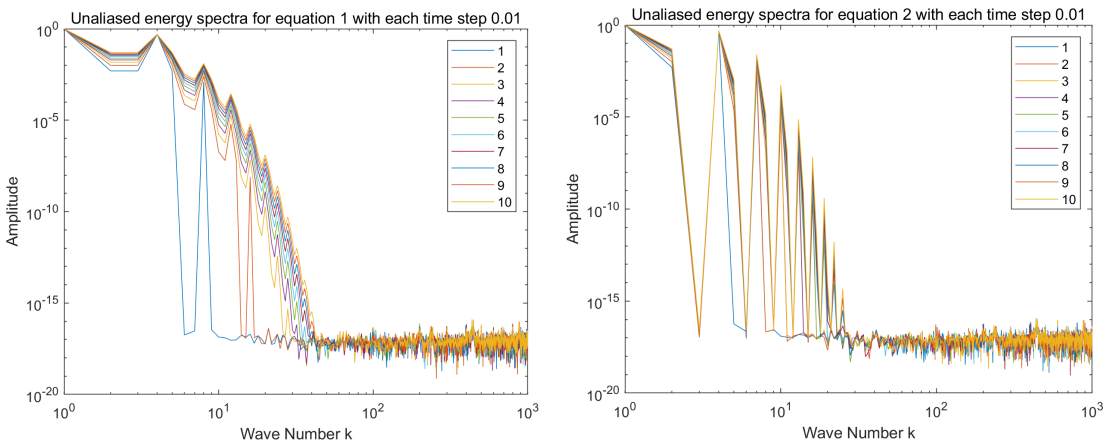
The energy spectra of the solutions of fully aliased scheme with time step $\Delta t = 0.1$ for both equations are plotted in Figure 9.




(a) (b)

Figure 9 Energy spectra of solutions of fully aliased scheme with time step $\Delta t = 0.1$ for equation 1(a) and equation 2(b)

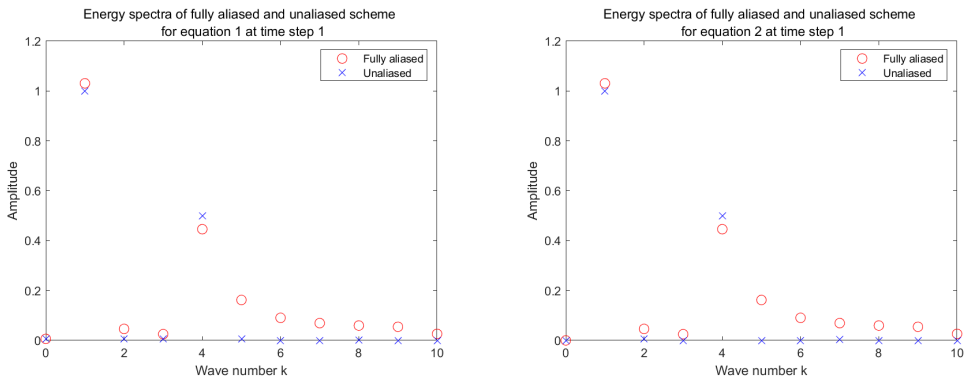
The energy spectra of the solutions of unaliased scheme with time step $\Delta t = 0.01$ for both equations are plotted in Figure 10.



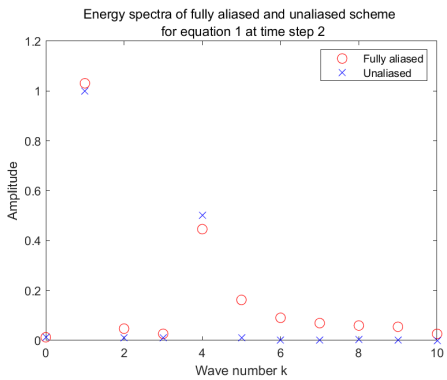

(a) (b)

Figure 10 Energy spectra of solutions of unaliased scheme with time step $\Delta t = 0.01$ for equation 1(a) and equation 2(b)

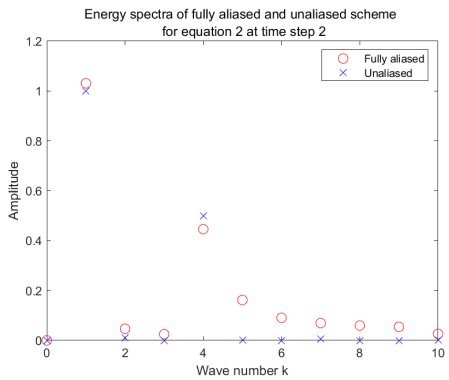
For both fully aliased and unaliased schemes, the energy spectra increase as the time step increases. To investigate the aliasing error, a comparative analysis of the energy spectra for both fully aliased and unaliased schemes is essential. This comparison focuses on the energy spectra at wave numbers 1 through 10, across time steps ranging from 1 to 10. Figure 11 illustrates these comparisons, showcasing the differences in energy spectra between the two schemes across the specified range of wave numbers and time steps.



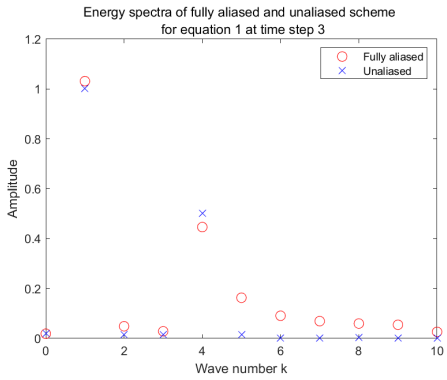

(a) (h)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 2
Fully aliased
Unaliased
Amplitude
Wave number k

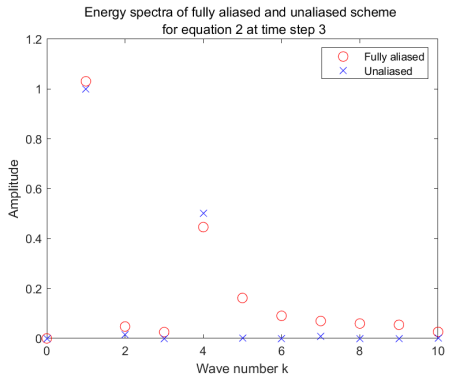
(b)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 2
Fully aliased
Unaliased
Amplitude
Wave number k

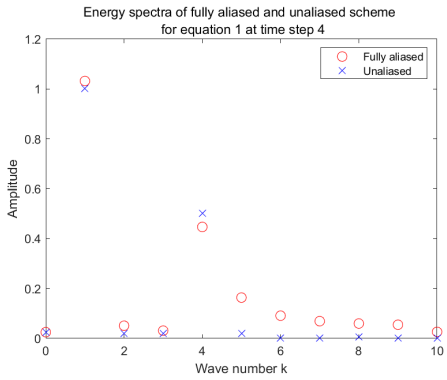
(i)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 3
Fully aliased
Unaliased
Amplitude
Wave number k

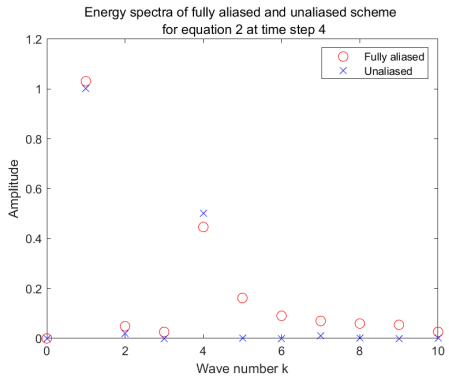
(c)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 3
Fully aliased
Unaliased
Amplitude
Wave number k

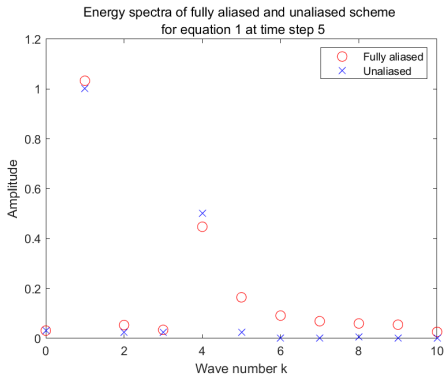
(j)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 4
Fully aliased
Unaliased
Amplitude
Wave number k

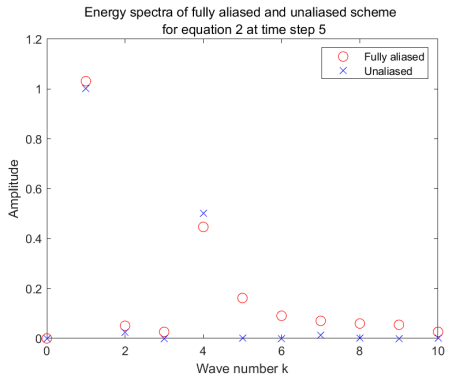
(d)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 4
Fully aliased
Unaliased
Amplitude
Wave number k

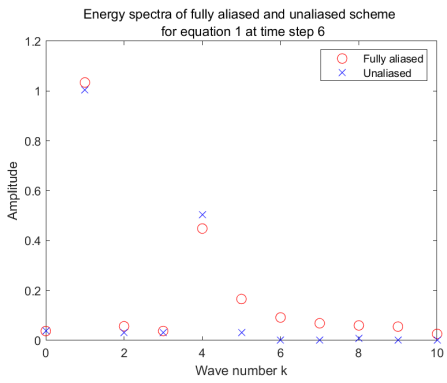
(k)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 5
Fully aliased
Unaliased
Amplitude
Wave number k

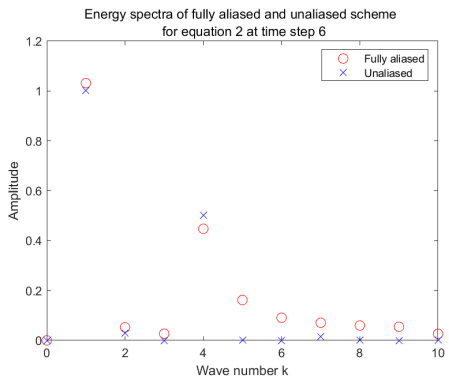
(e)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 5
Fully aliased
Unaliased
Amplitude
Wave number k

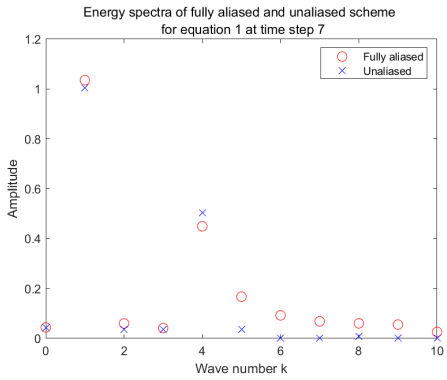
(l)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 6
Fully aliased
Unaliased
Amplitude
Wave number k

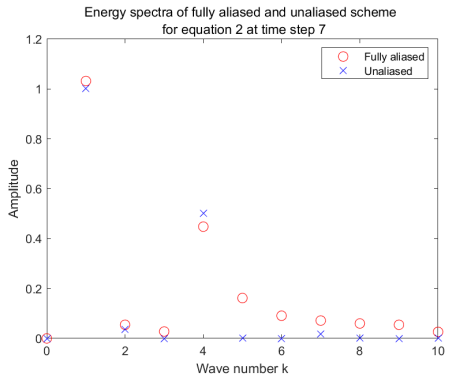
(f)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 6
Fully aliased
Unaliased
Amplitude
Wave number k

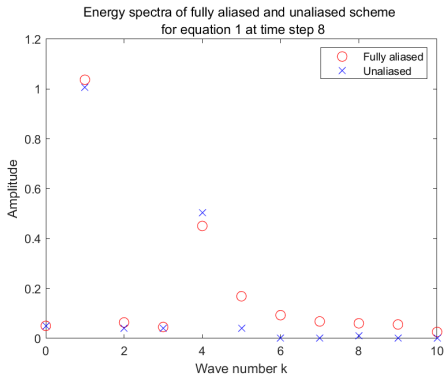
(m)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 7
Fully aliased
Unaliased
Amplitude
Wave number k

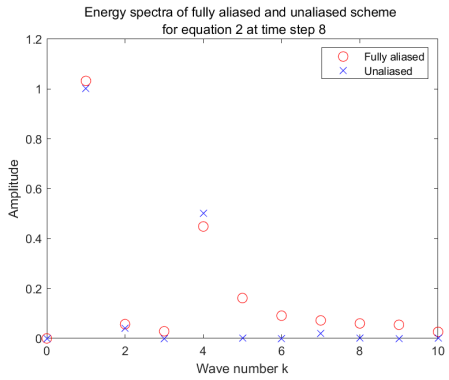
(g)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 7
Fully aliased
Unaliased
Amplitude
Wave number k

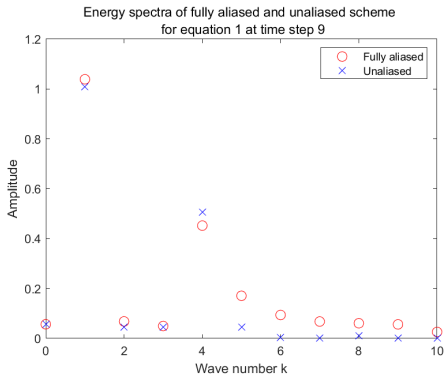
(n)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 8
Fully aliased
Unaliased
Amplitude
Wave number k

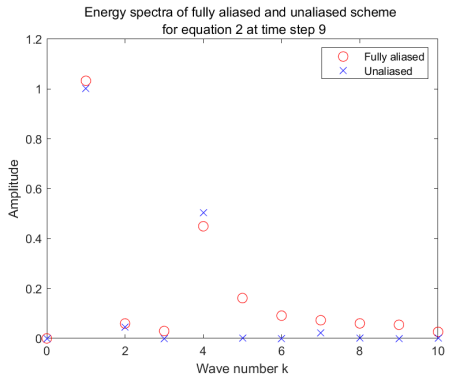
(h)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 8
Fully aliased
Unaliased
Amplitude
Wave number k

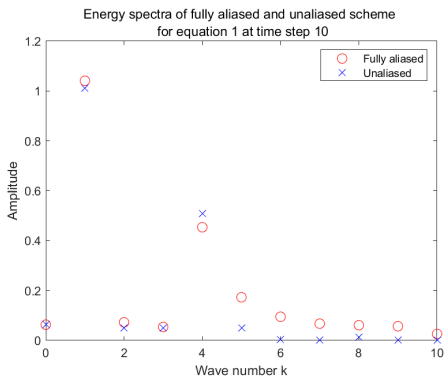
(o)


Energy spectra of fully aliased and unaliased scheme
for equation 1 at time step 9
Fully aliased
Unaliased
Amplitude
Wave number k

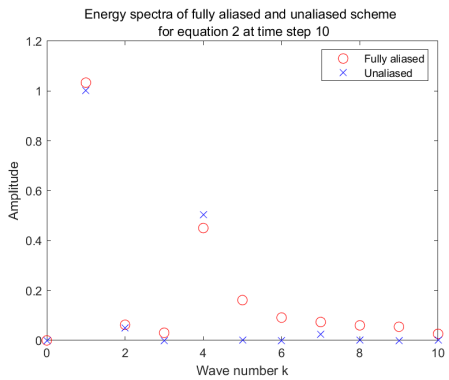
(i)


Energy spectra of fully aliased and unaliased scheme
for equation 2 at time step 9
Fully aliased
Unaliased
Amplitude
Wave number k

(p)




(j)




(q)

Figure 11 Energy spectra for fully aliased and unaliased schemes across wave numbers 1-10 and time steps 1-10

From Figure 11, it is evident that for both equation 1 and equation 2, the energy spectra of the fully aliased scheme are higher than those of the unaliased scheme at most time. This increase is due to aliasing error, a result of insufficient resolution where the amplitudes of higher wave numbers are erroneously added to those of lower wave numbers, artificially inflating the latter's amplitude. This phenomenon is clearly depicted in Figure 11. Furthermore, the difference in energy spectra between the fully aliased and unaliased schemes widens as the time step increases.

Upon comparing the results of equations 1 and 2, it is observable that the difference in energy spectra between the fully aliased and unaliased schemes for equation 2 is slightly smaller than that for equation 1. This is attributed to the non-linear term $u^2$ in equation 1 having a more significant impact on the increase of wave number compared to $\sin(x) \cdot u$, the non-linear term in equation 2.

### 2.4 Conclusion

This study examined the aliasing error in two time-dependent differential equations, finding that the fully aliased scheme consistently yielded higher energy spectra than the unaliased scheme due to aliasing error. This effect was more pronounced with an increase in time step size. Additionally, the difference in energy spectra was slightly less in Equation 2 compared to Equation 1, attributed to the differing impacts of their non-linear terms. These findings highlight the critical role of appropriate discretization in numerical simulations to minimize aliasing errors and achieve accurate solutions.

## 4. Appendix

MATLAB source code.

### 4.1 Problem 1

```
clear; clc;
```

```
%% Plot the solutions
clear;clc;
dx = 2*pi/20;
dt = 0.001;

figure;
m = 2;
T = 0.1;
xexa = linspace(0, 2*pi, 200);
uexa = sin(m*(xexa-T));
plot(xexa, uexa, 'k-', 'linewidth', 0.8)
hold on
order = 1;
[x, un] = Upwind(dt, dx, m, order, T);
plot(x, un', 'r-', 'linewidth', 0.8)
hold on
order = 2;
[x, un] = Upwind(dt, dx, m, order, T);
plot(x, un', 'b-', 'linewidth', 0.8)
hold on
order = 3;
[x, un] = Upwind(dt, dx, m, order, T);
plot(x, un', 'g-', 'linewidth', 0.8)
hold on
order = 4;
[x, un] = Upwind(dt, dx, m, order, T);
plot(x, un', '-', 'Color', [0.7, 0, 0.7],  'linewidth', 0.8)
xlabel('x')
ylabel('u')
titleStr = sprintf('(m = %s', num2str(m));
titleStr2 = sprintf('T = %s)', num2str(T));
title(['Solutions using upwind discretization schemes ' titleStr ', '
titleStr2]);
legend('Exact solution', '1^{st} order upwind', '2^{nd} order upwind',
'3^{rd} order upwind','3^{rd} order biased')

%% Modified wave numbers
beta = linspace(0, pi, 100);
dx =
2*pi/20;
k1_r = sin(beta)./dx;
k1_i = (cos(beta)-1)./dx;
figure;
plot(beta./pi, beta./dx, 'r-')
```

```
hold on
plot(beta./pi, k1_r)
hold on
plot(beta./pi, k1_i)

hold on
k2_r = (4.*sin(beta)-sin(2.*beta))./2./dx;
k2_i = (4.*cos(beta)-cos(2.*beta)-3)./2./dx;
plot(beta./pi, k2_r)
hold on
plot(beta./pi, k2_i)

hold on
k3_r = (18.*sin(beta)-9.*sin(2.*beta)+2.*sin(3.*beta))./6./dx;
k3_i = (18.*cos(beta)-9.*cos(2.*beta)+2.*cos(3.*beta)-11)./6./dx;
plot(beta./pi, k3_r)
hold on
plot(beta./pi, k3_i)

hold on
k3b_r = (8.*sin(beta)-sin(2.*beta))./6./dx;
k3b_i = (4.*cos(beta)-cos(2.*beta)-3)./6./dx;
plot(beta./pi, k3b_r)
hold on
plot(beta./pi, k3b_i)

x = linspace(0, 1, pi);
y = 0.*x;
hold on
plot(x,y,'k-')

xlabel('\beta/\pi (k\Deltax/\pi)')
ylabel('k''')
title('Modified wave numbers for different schemes');

%% Functions
function [x, un] = Upwind(dt, dx, m, order, T)
N=floor(2*pi/dx);
n=T/dt;

un=zeros(N+1,1);
for i = 1:N+1
    un(i) = sin(m*dx*(i-1));
end
```

```
if order == 1
    unew=zeros(N+2,1);
    un = [0, un'];
    un(1) = un(N+1);
    for i = 1:n
        for j = 2:N+2
            unew(j)=un(j)-dt/dx*(un(j)-un(j-1));
        end
        unew(1)=unew(N+1);
        un = unew;
    end
    x = linspace(0, 2*pi, N+1);
    un = un(2:end)';
end

if order == 2
    unew=zeros(N+3,1);
    un = [0, 0, un'];
    un(1) = un(N+1);
    un(2) = un(N+2);
    for i = 1:n
        for j = 3:N+3
            unew(j)=un(j)-dt/dx/2*(3*un(j)-4*un(j-1)+un(j-2));
        end
        unew(1)=unew(N+1);
        unew(2)=unew(N+2);
        un = unew;
    end
    x = linspace(0, 2*pi, N+1);
    un = un(3:end)';
end

if order == 3
    unew=zeros(N+4,1);
    un = [0, 0, 0, un'];
    un(1) = un(N+1);
    un(2) = un(N+2);
    un(3) = un(N+3);
    for i = 1:n
        for j = 4:N+4
            unew(j)=un(j)-dt/dx/6*(11*un(j)-18*un(j-1)+9*un(j-2)-2*un(j-3));
        end
        unew(1)=unew(N+1);
```

```
        unew(2)=unew(N+2);
        unew(3)=unew(N+3);
        un = unew;
    end
    x = linspace(0, 2*pi, N+1);
    un = un(4:end)';
end

if order == 4
    unew=zeros(N+4,1);
    un = [0, 0, un', 0];
    un(1) = un(N+1);
    un(2) = un(N+2);
    un(N+4) = un(4);
    for i = 1:n
        for j = 3:N+3
            unew(j)=un(j)-dt/dx/6*(2*un(j+1)+3*un(j)-6*un(j-1)+un(j-2));
        end
        unew(1)=unew(N+1);
        unew(2)=unew(N+2);
        unew(N+4)=unew(4);
        un = unew;
    end
    x = linspace(0, 2*pi, N+1);
    un = un(3:N+3)';
end

end
```

### 4.2 Problem 2

```
clear; clc;
%% Plot the energy spectra
dx = 2*pi/20;
dt = 0.1;
L = floor(2*pi/dx);
figure;

for i = 1:10
    t = i;
    un1_ali = ForwardEuler(dt, dx, 1, t);

    U = un1_ali;
    Y = fft(U);
    P2 = abs(Y/L);
```

```
    P1 = P2(1:L/2+1);
    P1(2:end-1) = 2*P1(2:end-1);
    f = (0:(L/2))/L * L;

    loglog(f, P1)
    hold on

end

title(['Fully aliased energy spectra for equation 1 with each time step ',
num2str(dt)]);
xlabel('Wave Number k')
ylabel('Amplitude')
legend('1','2','3','4','5','6','7','8','9','10')

%% Plot the fully aliased and unaliased energy spectra
dt = 0.01;
L = floor(2*pi/dx);
x = linspace(2, 2*pi, L+1);
colors = lines(10);

for i = 1:10
    t = i;
    dx = 2*pi/20;
    L = floor(2*pi/dx);
    un1_ali = ForwardEuler(dt, dx, 1, t);

    U = un1_ali;
    Y = fft(U);
    P2 = abs(Y/L);
    P1 = P2(1:L/2+1);
    P1(2:end-1) = 2*P1(2:end-1);
    P1_ali = zeros(11,1);
    P1_ali = P1;
    f = (0:(L/2))/L * L;

    figure;
    plot(f, P1_ali, 'ro', 'MarkerSize', 8);
    hold on

    dx = 2*pi/20/100;
    L = floor(2*pi/dx);
    un1_unali = ForwardEuler(dt, dx, 1, t);
    U = un1_unali;
```

```
    Y = fft(U);
    P2 = abs(Y/L);
    P1 = P2(1:L/2+1);
    P1(2:end-1) = 2*P1(2:end-1);
    P1_unali = zeros(11,1);
    P1_unali(:) = P1(1:11);

    plot(f, P1_unali, 'bx', 'MarkerSize', 8);
    title(['Energy spectra of fully aliased and unaliased scheme '
sprintf('\n') ' for equation 1 at time step ', num2str(t)], 'FontWeight',
'normal');
    xlabel('Wave number k')
    ylabel('Amplitude')
    legend('Fully aliased', 'Unaliased')

end

%% Functions
function [un] = ForwardEuler(dt, dx, eqnum, t)
N=floor(2*pi/dx);
n=t;

if eqnum == 1
    un=zeros(N+1,1);
    for i = 1:N+1
        x = (i-1)*dx;
        un(i) = sin(x) + 0.5*sin(4*x);
    end

    for i = 1:n
        for j = 1:N+1
            unew(j)=-un(j)^2*dt+un(j);
        end
        un = unew;
    end
    un = un';
end

if eqnum == 2
    un=zeros(N+1,1);
    for i = 1:N+1
        x = (i-1)*dx;
        un(i) = sin(x) + 0.5*sin(4*x);
    end
```

```
    for i = 1:n
        for j = 1:N+1
            x = (j-1)*dx;
            unew(j)=-sin(3*x)*un(j)*dt + un(j);
        end
        un = unew;
    end
    un = un';
end

end
```